

- **Route:** GET /api/v1/plants/

- **Headers:** `Authorization: Bearer <TOKEN>`
- **Behavior:**
- **GENERATOR** : Returns only facilities owned by the authenticated generator account.
- **REGULATOR** / **AUDITOR** / **ADMIN** : Returns all registered generation facilities market-wide.
- **Response (200 OK):** `json [{ "id": 1, "owner_id": 4, "name": "Mojave Desert Solar One", "fuel_type": "SOLAR", "nameplate_capacity_mw": 50.0, "grid_interconnection_id": "CAISO-SUB-9021", "location_address": "Mojave Desert, San Bernardino County, CA", "max_capacity_factor": 0.32, "status": "ACTIVE", "created_at": "2026-01-15T08:00:00Z" }]`

B. Register New Clean Power Facility

- **Route:** `POST /api/v1/plants/`
- **Headers:** `Authorization: Bearer <TOKEN>` (Roles: **GENERATOR** , **ADMIN**)
- **Request Body:** Facility registration payload.

3. Smart Meter Telemetry Ingestion (/meters)

A. Ingest Smart Meter Telemetry

- **Route:** `POST /api/v1/meters/`
- **Headers:** `Authorization: Bearer <TOKEN>`
- **Request Body:** `json { "plant_id": 1, "meter_serial_number": "MTR-SOL-001", "interval_start": "2026-03-01T00:00:00Z", "interval_end": "2026-03-31T23:59:59Z", "energy_generated_mwh": 1200.0, "raw_meter_hash": "a4f89d32 ... ", "telemetry_source": "UTILITY_SMART_METER" }`

4. Generation Claims & AI Fraud Detection (/claims)

A. Submit Claim & Trigger Real-Time AI Detection

- **Route:** `POST /api/v1/claims/`
- **Headers:** `Authorization: Bearer <TOKEN>` (Roles: **GENERATOR** , **ADMIN**)
- **Execution Pipeline:**
- Computes SHA-256 fingerprint of (`plant_id`, `period_start`, `period_end`, `claimed_mwh`).
- Runs Deterministic Rule Engine (`RULE-001` through `RULE-009`).
- Runs Scikit-learn Isolation Forest Anomaly Engine.
- Runs NetworkX Transfer Graph cycle check.
- Fuses scores into weighted composite score (0–100).
- Commits Claim Submission & Evaluation events onto Cryptographic Ledger.
- If verified (< 25.0), auto-mints digital REC certificate. If suspicious (≥ 65.0), sets status **HELD** and auto-opens regulatory case.
- **Request Body:** `json { "plant_id": 1, "period_start": "2026-03-01T00:00:00Z", "period_end": "2026-03-31T23:59:59Z", "claimed_mwh": 3800.0, "meter_reading_id": 1 }`
- **Response (201 Created):** `json { "id": 5, "claim_uid": "CLM-2026-FRAUD-MTR", "plant_id": 1, "claimed_mwh": 3800.0, "status": "HELD", "risk_score": 88.0, "risk_level": "CRITICAL",`

```
"risk_breakdown": { "rule_engine_score": 80.0, "ml_anomaly_score": 92.4, "graph_risk_score": 0.0,
"final_risk_score": 88.0, "risk_level": "CRITICAL", "recommendation": "HOLD",
"summary_explanation": "Flagged by 1 deterministic rules (METER_CLAIM_MISMATCH). Overclaimed by
216.7%. Machine learning model flagged statistical outlier (score: 92.4/100).", "factors": [ {
"rule_id": "RULE-005", "name": "METER_CLAIM_MISMATCH", "severity": "CRITICAL",
"score_contribution": 80.0, "description": "Claimed 3,800.0 MWh exceeds calibrated meter reading
of 1,200.0 MWh by 216.7% (Tolerance: 2.0%).", "flagged": true } ] } }
```

B. List Generation Claims

- **Route:** `GET /api/v1/claims/?status=HELD&limit=50`
- **Headers:** `Authorization: Bearer <TOKEN>`

5. REC Certificates, Wallet & Lineage Explorer (/certificates)

A. List Active REC Certificates

- **Route:** `GET /api/v1/certificates/`
- **Headers:** `Authorization: Bearer <TOKEN>`
- **Behavior:** Generators view only their owned digital wallet RECs; Regulators view all market RECs.

B. 6-Stage Lineage & Evidence Timeline Deep-Dive

- **Route:** `GET /api/v1/certificates/lineage/{identifier}`
- **Headers:** `Authorization: Bearer <TOKEN>`
- **Identifier:** Any Certificate UID (`REC-2026-SOL-09921`), Claim UID (`CLM-2026-FRAUD-MTR`), or Case Number (`CASE-2026-MTR-001`).
- **Response (200 OK):**

```
json { "found": true, "query": "CLM-2026-FRAUD-MTR", "search_type": "CLAIM",
"plant": { "name": "Mojave Desert Solar One", "capacity_mw": 50.0 }, "meter": { "serial": "MTR-
SOL-001", "energy_generated_mwh": 1200.0 }, "claim": { "uid": "CLM-2026-FRAUD-MTR", "mwh":
3800.0, "status": "HELD", "risk_score": 88.0 }, "investigation": { "case_number": "CASE-2026-MTR-
001", "priority": "CRITICAL", "status": "OPEN" }, "certificate": null, "transfers": [],
"ledger_blocks": [ { "index": 2, "event_type": "CLAIM_SUBMITTED", "current_hash": "b8f2 ... " }, {
"index": 3, "event_type": "CLAIM_EVALUATED", "current_hash": "c9a1 ... " } ], "timeline": [ {
"timestamp": "2026-03-01T00:00:00Z", "stage": "GROUND_TRUTH", "title": "Calibrated Meter Reading
Recorded", "description": "Meter MTR-SOL-001 recorded 1,200.0 MWh generated at Mojave Desert
Solar One.", "severity": "INFO", "actor": "Smart Meter IoT" }, { "timestamp": "2026-03-
12T10:00:00Z", "stage": "SUBMISSION", "title": "Generation Claim Submitted (CLM-2026-FRAUD-MTR)",
"description": "Submitted 3,800.0 MWh for period Mar 01, 2026 to Mar 31, 2026.", "severity":
"INFO", "actor": "Clean Energy Producer" }, { "timestamp": "2026-03-12T10:00:01Z", "stage":
"AI_DETECTION", "title": "Multi-Engine Forensic Evaluation: 88.0/100", "description":
"Recommendation: HOLD. Flagged by METER_CLAIM_MISMATCH (+216.7%).", "severity": "CRITICAL",
"actor": "AI Forensic Engine" }, { "timestamp": "2026-03-12T10:00:02Z", "stage": "INVESTIGATION",
"title": "Regulatory Investigation Case Opened (CASE-2026-MTR-001)", "description": "Priority:
CRITICAL. Status: OPEN.", "severity": "CRITICAL", "actor": "Regulatory Officer" } ] }
```

C. Peer-to-Peer Certificate Transfer

- **Route:** `POST /api/v1/certificates/{cert_id}/transfer`
- **Headers:** `Authorization: Bearer <TOKEN>`
- **Terminal Guard:** Rejects with `HTTP 400` if status is `REDEEMED` or `REVOKED`.

D. Scope 2 Compliance Retirement / Redemption

- **Route:** `POST /api/v1/certificates/{cert_id}/redeem`
- **Headers:** `Authorization: Bearer <TOKEN>`
- **Terminal Action:** Permanently marks certificate `REDEEMED` and commits proof block to the ledger.

6. Regulatory Investigations & Human-in-the-Loop Adjudication (`/investigations`)

A. List Open Investigation Cases

- **Route:** `GET /api/v1/investigations/?status=OPEN`
- **Headers:** `Authorization: Bearer <TOKEN>` (Roles: `REGULATOR`, `AUDITOR`, `ADMIN`)

B. Adjudicate Case (Sign & Seal Judicial Decision)

- **Route:** `POST /api/v1/investigations/{case_id}/decision`
- **Headers:** `Authorization: Bearer <TOKEN>` (Strict RBAC: `REGULATOR`, `ADMIN` only; Generators blocked with `HTTP 403`)
- **Request Body:** `json { "decision_action": "CONFIRM_FRAUD_HOLD", "findings": "Audit confirms meter reading bypass. Overclaim exceeds physical reality. Statutory fine issued." }`
- **Alternate Action:** `"decision_action": "CLEAR_AND_ISSUE"` clears false positive, approves claim, and mints certificate onto ledger.

7. Cryptographic Trust Ledger & Tamper Detection (`/ledger`)

A. Ledger Chain Integrity Audit

- **Route:** `GET /api/v1/ledger/verify`
- **Headers:** `Authorization: Bearer <TOKEN>`
- **Response (`200 OK`):** `json { "is_valid": true, "total_blocks": 28, "tampered_block_index": null, "verification_message": "Ledger integrity confirmed: All 28 cryptographic blocks are valid and un-tampered.", "verified_at": "2026-09-12T16:30:00Z" }`

B. Drag-and-Drop Document Hash Tamper Verifier

- **Route:** `POST /api/v1/ledger/verify-document`
 - **Form Data:** `file: <FILE_BLOB>`
 - **Response:** Computes SHA-256 in memory and matches against registry database.
-

8. Analytics & Graph Surveillance (/analytics)

A. Role-Scoped Dashboard KPIs

- **Route:** `GET /api/v1/analytics/dashboard`
- **Headers:** `Authorization: Bearer <TOKEN>`
- **Role Isolation:** Scoped to owned assets for Generators; market-wide macro surveillance for Regulators.

B. Secondary Market Network Graph & Cycle Detection

- **Route:** `GET /api/v1/analytics/network-graph`
- **Headers:** `Authorization: Bearer <TOKEN>`
- **Response:**

```
json { "nodes": [ { "id": "user-4", "label": "Helios Solar", "role": "GENERATOR" }, { "id": "user-5", "label": "Boreas Wind", "role": "GENERATOR" }, { "id": "user-6", "label": "Global Carbon", "role": "GENERATOR" } ], "edges": [ { "from": "user-5", "to": "user-6", "volume": 5000.0 }, { "from": "user-6", "to": "user-4", "volume": 5000.0 }, { "from": "user-4", "to": "user-5", "volume": 5000.0 } ], "detected_cycles": [ ["user-5", "user-6", "user-4", "user-5"] ], "clustering_coefficient": 0.42 }
```